INCH-POUND
MIL-C-24640/5C
26 June 1995
SUPERSEDING
MIL-C-24640/5B
31 July 1989

MILITARY SPECIFICATION SHEET

* CABLE, ELECTRIC, 600 VOLTS, LIGHT WEIGHT,
MULTI PAIR AWG 20, TYPE TTXS
(INCLUDING VARIATION TYPE TTXSA AND TTXSO)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.

CONSTRUCTION
(Non-Watertight)

FIRST - 19 strand copper conductor in accordance with ASTM B-286, tinned strands in accordance with ASTM B-33. AWG size 20. Maximum diameter 0.042 inches

SECOND - Component wires shall be MIL-W-81044/12 and shall be tested in accordance with the component wire ratings and requirements listed on this slash sheet. Quality conformance testing shall be in accordance with Table VI of MIL-C-24640.

THIRD - Two component wires, one black and one white, shall be twisted together to form the required number of pairs. The pitch shall be 1 " ± ¼" (inch).

FOURTH - Binder tape applied helically with overlap, at manufacturers option.

FIFTH - Braided shield of # 38 AWG tin coated copper strands. Braid coverage shall be no less than 90% determined in accordance with MIL-C-27500.

SIXTH - Shield insulation of two polyester tapes or polyester tape and extruded jacket. Identification codes applied using Method 2.

AMSC N/A FSC 6145

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CONSTRUCTION
(Non-Watertight)
(continued)

SEVENTH - The required number of shielded, twisted pairs shall be cabled together. Cabling sequence shall be consecutive, starting with shielded single number 1 from center outward. Fillers may be used as necessary, to form a round cable.

EIGHTH - Binder tape applied helically with overlap.

CONSTRUCTION VARIANT FOR TYPE TTXS

NINTH - Crosslinked polyolefin jacket. (See Table I for thickness).

CONSTRUCTION VARIANT FOR TYPE TTXSA

NINTH - Crosslinked polyolefin jacket. (See Table II for thickness).

TENTH - Braided metal armor. Dimensions in accordance with Table II.

CONSTRUCTION VARIANT FOR TYPE TTXSO

NINTH - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface Transfer Impedance requirements as listed herein.

TENTH - Binder tape applied helically with overlap, at manufacturer's option.

ELEVENTH Crosslinked polyolefin jacket. (See Table III for thickness).

TABLE I. Construction details - unarmored.

PIN M24640/5	Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch) min	Cable diameter (inch) max	Cable weight (pounds per 1000 feet) (max)
-01UN	TTXS-2	2	0.038	0.335	0.361	74.3
-02UN	TTXS-4	4	.041	.393	.424	115.0

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TABLE II. Construction details - armored.

PIN M24640/5	Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch) min	Cable diameter (inch) max	Cable weight (pounds per 1000 feet) (max)
-01AN	TTXSA-2	2	0.038	0.385	0.411	100.2
-02AN	TTXSA-4	4	.041	.443	.474	150.4

Table III. Construction details - overall shield.

PIN M24640/5	Type number	Number of pairs	Diameter over shield (inch) (nom)	Jacket thickness (inch) (minimum	Cable diameter (inch) min	Cable diameter (inch) max	Cable weight (pounds per 1000 feet) (max)
-01UO	TTXSO-2	2	0.283	0.041	0.386	0.416	126.0
-02UO	TTXSO-6	6	.418	.044	.506	.546	219.0
-03UO	TTXSO-8	8	.510	.044	.587	.633	284.0
-04UO	TTXSO-10	10	.553	.044	.627	.675	322.0

REQUIREMENTS:

Qualification required.
Construction details: See tables I, II and III.

Component wire:

For the component wire the following additional tests shall be applicable in accordance with MIL-W-81044:

WIRE, RATINGS AND ADDITIONAL REQUIREMENTS:

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 300± 2°C (572± 3.6°F) for 6 hours.
for 6 hours.
Blocking: 150± 2°C (302± 3.6°F)
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test:
Primary insulation (when tested in lieu of spark test): 6.0 kilovolts (peak) 100 percent test.
Component wire: 8.0 kilovolts (peak), or 3 Khz spark test, 5.5 KV, 100 percent test.

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Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200± 2°C (392± 3.6°F) for 168 hours.
Physical properties of primary insulation:
Tensile strength, 2500 lb/in^2 (min).
Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 300± 2°C (572± 3.6°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz, 100 percent test.
Surface resistance: 500 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150± 2°C (302± 3.6°F).
Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	10.95
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor and shield	1000
Conductor to shield	Not applicable
Shield to shield (component shield only)	200
Shield to shield (component shield to overall shield (TTXSO only))	500
Insulation resistance - megohms/1000 feet, minimum	
Conductor to conductor and shield	2500
Shield to shield (component shield only)	100
Shield to shield (component shield to overall shield (TTXSO only))	100
Conductor and shield continuity	No failure
Jacket flaws (TTXSO only)	No failure
Group A:	
Visual and dimensional	No failure
Capacitance:	
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	50
Capacitance unbalance - percent, maximum	15
Characteristic impedance - at 1 MHz, ohms	30 min, 63 max
Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	Not applicable
Weight	No failure
Group B:	
Cold bending	No failure
* Crosslink proof test (percent maximum)	
Jacket	50
* Drip	Not applicable
Tear - pounds per inch thickness, minimum	35

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	Physicals (unaged)	
	Jacket	
	Tensile strength - lb/in^2, minimum	1300
	Elongation - percent, minimum	160
	Attenuation - at 3 MHz, dB/100 feet, maximum	3
	Gas flame (1 hour)	Not applicable
	Group C:	
	Physicals (aged)	
	Jacket	
	Air oven	
	Tensile strength - percent of unaged, minimum	60
	Elongation - percent of unaged, minimum	60
	Hot oil immersion	
*	Tensile strength - percent of unaged minimum	50
*	Elongation - percent of unaged, minimum	50
*	Permanence of printing (conductor) - cycles, minimum	Not applicable
*	Permanence of printing (jacket) - cycles, minimum	125
	Armor - (type TTXSA only) - conformance to material, construction and coverage	No failure
*	Shield - conformance to material, construction and coverage	No failure
	Heat distortion - percent of unaged, maximum	30
*	Cable sealant removability	Not applicable
	Shrinkage	No failure
	Mutual inductance - dB/1000 feet, maximum	Not applicable
*	Surface transfer impedance (TTXSO only)	
	Milliohms per meter, maximum	700
	EMP response, dB, minimum	60
	Pulse response time - ms/500 feet	
	Rise time, maximum	Not applicable
	Decay time, maximum	Not applicable
*	Group D:	
	Flame propagation	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility cable	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18

QUALIFICATION INSPECTION (continued):

Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50

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*	Smoke index, maximum	
	Jacket	25
	Fillers	45
	Insulation	10
*	Toxicity index, maximum	
	Jacket	5
	Fillers	5
	Insulation	1.5
	Durometer (jacket) - (type A) hardness, minimum	80
	Weathering (jacket)	No failure
	Fungus Resistance	No failure
	Abrasion resistance	No failure
	Gas Flame (3 hour)	Not applicable
*	Component wire	No failure

Unit ordering length: 250 feet nominal.

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH

Review Activities:
Army - AV, CR, ER, ME, AR, AL
Navy - EC, CG
DLA - IS

Agent:
DLA-IS

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